Inductive Moment Matching

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Figure 1. Generated samples on ImageNet-256×256 using 8 steps.

Abstract

Diffusion models and Flow Matching generate high-quality samples but are slow at inference, and distilling them into few-step models often leads to instability and extensive tuning. To resolve these trade-offs, we propose Inductive Moment Matching (IMM), a new class of generative models for one- or few-step sampling with a *single-stage* training procedure. Unlike distillation, IMM does not require pre-training initialization and optimization of two networks; and unlike Consistency Models, IMM guarantees distribution-level convergence and remains stable under various hyperparameters and standard model architectures. IMM surpasses diffusion models on ImageNet-256×256 with 1.99 FID using only 8 inference steps and achieves state-of-the-art 2-step FID of 1.98 on CIFAR-10 for a model trained from scratch.

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1. Introduction

Generative models for continuous domains have enabled numerous applications in images (Rombach et al., 2022; Saharia et al., 2022; Esser et al., 2024), videos (Ho et al., 2022a; Blattmann et al., 2023; OpenAI, 2024), and audio (Chen et al., 2020; Kong et al., 2020; Liu et al., 2023), yet achieving high-fidelity outputs, efficient inference, and stable training remains a core challenge — a trilemma that continues to motivate research in this domain. Diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2020b), one of the leading techniques, require many inference steps for high-quality results, while step-reduction methods, such as diffusion distillation (Yin et al., 2024; Sauer et al., 2025; Zhou et al., 2024; Luo et al., 2024a) and Consistency Models (Song et al., 2023; Geng et al., 2024; Lu & Song, 2024; Kim et al., 2023), often risk training collapse without careful tuning and regularization (such as pre-generating data-noise pair and early stopping).

To address the aforementioned trilemma, we introduce Inductive Moment Matching (IMM), a stable, *single-stage* training procedure that learns generative models from scratch for single- or multi-step inference. IMM operates on the time-dependent marginal distributions of stochastic interpolants (Albergo et al., 2023) — continuous-time stochastic processes that connect two arbitrary probability density functions (data at $t = 0$ and prior at $t = 1$). By learning a (stochastic or deterministic) mapping from any marginal at time t to any marginal at time $s < t$, it can naturally support one- or multi-step generation (Figure 2).

IMM models can be trained efficiently from mathematical induction. For time $s < r < t$, we form two distributions at s by running a one-step IMM from samples at r and t. We then minimize their divergence, enforcing that the distributions at s are independent of the starting time-steps. This construction by induction guarantees convergence to the data distribution. To help with training stability, we model IMM based on certain stochastic interpolants and optimize the objective with stable sample-based divergence estimators such as moment matching (Gretton et al., 2012). Notably, we prove that Consistency Models (CMs) are a single-particle, first-moment matching special case of IMM, which partially explains the training instability of CMs.

On ImageNet-256×256, IMM surpasses diffusion models and achieves 1.99 FID with only 8 inference steps using standard transformer architectures. On CIFAR-10, IMM similarly achieves state-of-the-art of 1.98 FID with 2-step generation for a model trained from scratch.

2. Preliminaries

2.1. Diffusion, Flow Matching, and Interpolants

For a data distribution $q(\mathbf{x})$, Variance-Preserving (VP) diffusion models (Ho et al., 2020; Song et al., 2020b) and Flow Matching (FM) (Lipman et al., 2022; Liu et al., 2022) construct time-augmented variables $\mathbf{x}_t$ as an interpolation between data $\mathbf{x} \sim q(\mathbf{x})$ and prior $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$ such that $\mathbf{x}_t = \alpha_t \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$ where $\alpha_0 = \sigma_1 = 1, \alpha_1 = \sigma_0 = 0$. VP diffusion commonly chooses $\alpha_t = \cos(\frac{\pi}{2}t), \sigma_t = \sin(\frac{\pi}{2}t)$ and FM chooses $\alpha_t = 1 - t, \sigma_t = t$. Both v-prediction diffusion (Salimans & Ho, 2022) and FM are trained by matching the conditional velocity $\mathbf{v}_t = \alpha_t' \mathbf{x} + \sigma_t' \boldsymbol{\epsilon}$ such that a neural network $\boldsymbol{G}_\theta(\mathbf{x}_t, t)$ approximates $\mathbb{E}_{\mathbf{x},\boldsymbol{\epsilon}}[\mathbf{v}_t | \mathbf{x}_t]$. Samples can then be generated via probability-flow ODE (PF-ODE) $\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \boldsymbol{G}_\theta(\mathbf{x}_t, t)$ starting from $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$.

Stochastic interpolants. Unifying diffusion models and FM, stochastic interpolants (Albergo et al., 2023; Albergo & Vanden-Eijnden, 2022) construct a conditional interpolation $q_t(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon}) = \mathcal{N}(\boldsymbol{I}_t(\mathbf{x}, \boldsymbol{\epsilon}), \gamma_t^2 I)$ between any data $\mathbf{x} \sim q(\mathbf{x})$ and prior $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$ and sets constraints $\boldsymbol{I}_1(\mathbf{x}, \boldsymbol{\epsilon}) = \boldsymbol{\epsilon}$, $\boldsymbol{I}_0(\mathbf{x}, \boldsymbol{\epsilon}) = \mathbf{x}$, and $\gamma_1 = \gamma_0 = 0$. Similar to FM, a deterministic sampler can be learned by explicitly matching the conditional interpolant velocity $\mathbf{v}_t = \partial_t \boldsymbol{I}_t(\mathbf{x}, \boldsymbol{\epsilon}) + \dot{\gamma}_t \mathbf{z}$ where $\mathbf{z} \sim \mathcal{N}(0, I)$ such that $\boldsymbol{G}_\theta(\mathbf{x}_t, t) \approx \mathbb{E}_{\mathbf{x},\boldsymbol{\epsilon},\mathbf{z}}[\mathbf{v}_t|\mathbf{x}_t]$. Sampling is performed following the PF-ODE $\frac{\mathrm{d}\mathbf{x}_t}{\mathrm{d}t} = \boldsymbol{G}_\theta(\mathbf{x}_t, t)$ similarly starting from prior $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$.

When $\gamma_t \equiv 0$ and $\boldsymbol{I}_t(\mathbf{x}, \boldsymbol{\epsilon}) = \alpha_t \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$ for α_t, σ_t defined in FM, the intermediate variable $\mathbf{x}_t = \alpha_t \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$ becomes a deterministic interpolation and its interpolant velocity $\mathbf{v}_t = \alpha_t' \mathbf{x} + \sigma_t' \boldsymbol{\epsilon}$ reduces to FM velocity. Thus, its training and inference both reduce to that of FM. When $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, stochastic interpolants reduce to v-prediction diffusion.

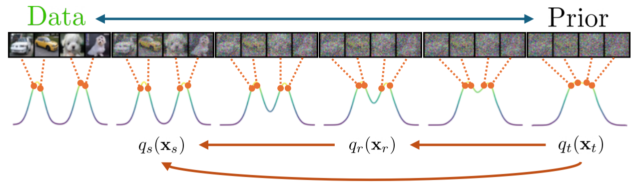

Figure 2. Using an interpolation from data to prior, we define a one-step sampler that moves from any t to $s < t$, directly transforming $q_t(\mathbf{x}_t)$ to $q_s(\mathbf{x}_s)$. This can be repeated by jumping to an intermediate $r < t$ before moving to $s < r$.

2.2. Maximum Mean Discrepancy

Maximum Mean Discrepancy (MMD, Gretton et al. (2012)) between distribution $p(\mathbf{x}), q(\mathbf{y})$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$ is an integral probability metric (Müller, 1997) commonly defined on Reproducing Kernel Hilbert Space (RKHS) $\mathcal{H}$ with a positive definite kernel $k : \mathbb{R}^D \times \mathbb{R}^D \to \mathbb{R}$ as

$$\mathrm{MMD}^2(p(\mathbf{x}), q(\mathbf{y})) = \|\mathbb{E}_{\mathbf{x}}[k(\mathbf{x}, \cdot)] - \mathbb{E}_{\mathbf{y}}[k(\mathbf{y}, \cdot)]\|_{\mathcal{H}}^2 \quad (1)$$

where the norm is in $\mathcal{H}$. Choices such as the RBF kernel imply an inner product of infinite-dimensional feature maps consisting of *all* moments of $p(\mathbf{x})$ and $q(\mathbf{y})$, *i.e.* $\mathbb{E}[\mathbf{x}^j]$ and $\mathbb{E}[\mathbf{y}^j]$ for integer $j \geq 1$ (Steinwart & Christmann, 2008).

3. Inductive Moment Matching

We introduce Inductive Moment Matching (IMM), a method that trains a model of both high quality and sampling efficiency in a single stage. To do so, we assume a time-augmented interpolation between data (distribution at $t = 0$) and prior (distribution at $t = 1$) and propose learning an implicit one-step model (*i.e.* a one-step sampler) that transforms the distribution at time t to the distribution at time s for any $s < t$ (Section 3.1). The model enables direct one-step sampling from $t = 1$ to $s = 0$ and few-step sampling via recursive application from any t to any $r < t$ and then to any $s < r$ until $s = 0$; this allows us to learn the model from its own samples via bootstrapping (Section 3.2).

3.1. Model Construction via Interpolants

Given data $\mathbf{x} \sim q(\mathbf{x})$ and prior $\boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon})$, the time-augmented interpolation $\mathbf{x}_t$ defined in Albergo et al. (2023) follows $\mathbf{x}_t \sim q_t(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon})$. This implies a marginal interpolating distribution

$$q_t(\mathbf{x}_t) = \iint q_t(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon}) q(\mathbf{x}) p(\boldsymbol{\epsilon}) \mathrm{d}\mathbf{x} \mathrm{d}\boldsymbol{\epsilon}. \quad (2)$$

We learn a model distribution implicitly defined by a one-step sampler that transforms $q_t(\mathbf{x}_t)$ into $q_s(\mathbf{x}_s)$ for some $s \le t$. This can be done via a special class of interpolants, which preserves the marginal distribution $q_s(\mathbf{x}_s)$ while interpolating between $\mathbf{x}$ and $\mathbf{x}_t$. We term these *marginal-preserving* interpolants among a class of generalized interpolants. Formally, we define $\mathbf{x}_s$ as a generalized interpolant between $\mathbf{x}$ and $\mathbf{x}_t$ if, for all $s \in [0, t]$, its distribution follows

$$q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) = \mathcal{N}(\boldsymbol{I}_{s|t}(\mathbf{x}, \mathbf{x}_t), \gamma^2_{s|t} I) \qquad (3)$$

and satisfies constraints $\boldsymbol{I}_{t|t}(\mathbf{x}, \mathbf{x}_t) = \mathbf{x}_t$, $\boldsymbol{I}_{0|t}(\mathbf{x}, \mathbf{x}_t) = \mathbf{x}$, $\gamma_{t|t} = \gamma_{0|t} = 0$, and $q_{t|1}(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon}) \equiv q_t(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon})$. When $t = 1$, it reduces to regular stochastic interpolants. Next, we define marginal-preserving interpolants.

Definition 1 (Marginal-Preserving Interpolants)**.** A generalized interpolant $\mathbf{x}_s$ is *marginal-preserving* if for all $t \in [0, 1]$ and for all $s \in [0, t]$, the following equality holds:

$$q_s(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) q_t(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) \mathrm{d}\mathbf{x}_t \mathrm{d}\mathbf{x}, \qquad (4)$$

where

$$q_t(\mathbf{x}|\mathbf{x}_t) = \int \frac{q_t(\mathbf{x}_t|\mathbf{x}, \boldsymbol{\epsilon}) q(\mathbf{x}) p(\boldsymbol{\epsilon})}{q_t(\mathbf{x}_t)} \mathrm{d}\boldsymbol{\epsilon}. \qquad (5)$$

That is, this class of interpolants has the same marginal at s regardless of t. For all $t \in [0, 1]$, we define our *noisy* model distribution at $s \in [0, t]$ as

$$p^\theta_{s|t}(\mathbf{x}_s) = \iint q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) p^\theta_{s|t}(\mathbf{x}|\mathbf{x}_t) q_t(\mathbf{x}_t) \mathrm{d}\mathbf{x}_t \mathrm{d}\mathbf{x} \qquad (6)$$

where the interpolant is marginal preserving and $p^\theta_{s|t}(\mathbf{x}|\mathbf{x}_t)$ is our *clean* model distribution implicitly parameterized as a one-step sampler. This definition also enables multi-step sampling. To produce a clean sample $\mathbf{x}$ given $\mathbf{x}_t \sim q_t(\mathbf{x}_t)$ in two steps via an intermediate s: (1) we sample $\hat{\mathbf{x}} \sim p^\theta_{s|t}(\mathbf{x}|\mathbf{x}_t)$ followed by $\hat{\mathbf{x}}_s \sim q_{s|t}(\mathbf{x}_s|\hat{\mathbf{x}}, \mathbf{x}_t)$ and (2) if the marginal of $\hat{\mathbf{x}}_s$ matches $q_s(\mathbf{x}_s)$, we can obtain $\mathbf{x}$ by $\mathbf{x} \sim p^\theta_{0|s}(\mathbf{x}|\hat{\mathbf{x}}_s)$. We are therefore motivated to minimize divergence between Eq. (4) and (6) using the objective below.

Naïve objective. As one can easily draw samples from the model, it can be naïvely learned by directly minimizing

$$\mathcal{L}(\theta) = \mathbb{E}_{s,t}\left[D(q_s(\mathbf{x}_s), p^\theta_{s|t}(\mathbf{x}_s))\right] \qquad (7)$$

with time distribution $p(s, t)$ and a sample-based divergence metric $D(\cdot, \cdot)$ such as MMD or GAN (Goodfellow et al., 2020). If an interpolant $\mathbf{x}_s$ is marginal-preserving, then the minimum loss is 0 (see Lemma 3). One might also notice the similarity between right-hand sides of Eq. (4) and (6). However, $q_s(\mathbf{x}_s) = p^\theta_{s|t}(\mathbf{x}_s)$ does not necessarily imply $p^\theta_{s|t}(\mathbf{x}|\mathbf{x}_t) = q_t(\mathbf{x}|\mathbf{x}_t)$. In fact, the minimizer $p^\theta_{s|t}(\mathbf{x}|\mathbf{x}_t)$ is not unique and, under mild assumptions, a deterministic minimizer exists (see Section 4).

3.2. Learning via Inductive Bootstrapping

While sound, the naïve objective in Eq. (7) is difficult to optimize in practice because when t is far from s, the input distribution $q_t(\mathbf{x}_t)$ can be far from the target $q_s(\mathbf{x}_s)$. Fortunately, our interpolant construction implies that the model definition in Eq. (6) satisfies boundary condition $q_s(\mathbf{x}_s) = p^\theta_{s|s}(\mathbf{x}_s)$ regardless of θ (see Lemma 4), which indicates that $p^\theta_{s|t}(\mathbf{x}_s) \approx q_s(\mathbf{x}_s)$ when t is close to s. Furthermore, the interpolant enforces $p^\theta_{s|t}(\mathbf{x}_s) \approx p^\theta_{s|r}(\mathbf{x}_s)$ for any $r < t$ close to t as long as the model is continuous around t. Therefore, we can construct an inductive learning algorithm for $p^\theta_{s|t}(\mathbf{x}_s)$ by using samples from $p^\theta_{s|r}(\mathbf{x}_s)$.

For better analysis, we define a sequence number n for parameter θ_n and function $r(s, t)$ where $s \le r(s, t) < t$ such that $p^{\theta_n}_{s|t}(\mathbf{x}_s)$ learns to match $p^{\theta_{n-1}}_{s|r}(\mathbf{x}_s)$.[1] We omit r's arguments when context is clear and let $r(s, t)$ be a finite decrement from t but truncated at $s \le t$ (see Appendix B.3 for well-conditioned $r(s, t)$).

General objective. With marginal-preserving interpolants and mapping $r(s, t)$, we learn θ_n in the following objective:

$$\mathcal{L}(\theta_n) = \mathbb{E}_{s,t}\left[w(s, t)\mathrm{MMD}^2(p^{\theta_{n-1}}_{s|r}(\mathbf{x}_s), p^{\theta_n}_{s|t}(\mathbf{x}_s))\right] \qquad (8)$$

where $w(s, t)$ is a weighting function. We choose MMD as our objective due to its superior optimization stability and show that this objective learns the correct data distribution.

Theorem 1. *Assuming $r(s, t)$ is well-conditioned, the interpolant is marginal-preserving, and θ^*_n is a minimizer of Eq.* (8) *for each n with infinite data and network capacity, for all $t \in [0, 1]$, $s \in [0, t]$,*

$$\lim_{n\to\infty} \mathrm{MMD}^2(q_s(\mathbf{x}_s), p^{\theta^*_n}_{s|t}(\mathbf{x}_s)) = 0. \qquad (9)$$

In other words, θ_n eventually learns the target distribution $q_s(\mathbf{x}_s)$ by parameterizing a one-step sampler $p^{\theta_n}_{s|t}(\mathbf{x}|\mathbf{x}_t)$.

4. Simplified Formulation and Practice

We present algorithmic and practical decisions below.

4.1. Algorithmic Considerations

Despite theoretical soundness, it remains unclear how to empirically choose a marginal-preserving interpolant. First, we present a sufficient condition for marginal preservation.

Definition 2 (Self-Consistent Interpolants)**.** Given $s, t \in [0, 1]$, $s \le t$, an interpolant $\mathbf{x}_s \sim q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t)$ is *self-consistent* if for all $r \in [s, t]$, the following holds:

$$q_{s|t}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_t) = \int q_{s|r}(\mathbf{x}_s|\mathbf{x}, \mathbf{x}_r) q_{r|t}(\mathbf{x}_r|\mathbf{x}, \mathbf{x}_t) \mathrm{d}\mathbf{x}_r \qquad (10)$$

[1] Note that n is different from optimization steps. Advancing from $n-1$ to n can take arbitrary number of optimization steps.